

Battery Degradation and State-of-Health Digital Twin

Electro-thermal simulation, measured SOH uncertainty, and early-cycle RUL evidence.

Electro-thermal

Cell and pack response with variability

Aging and RUL

Calendar, cycle, SOH, and EOL workflows

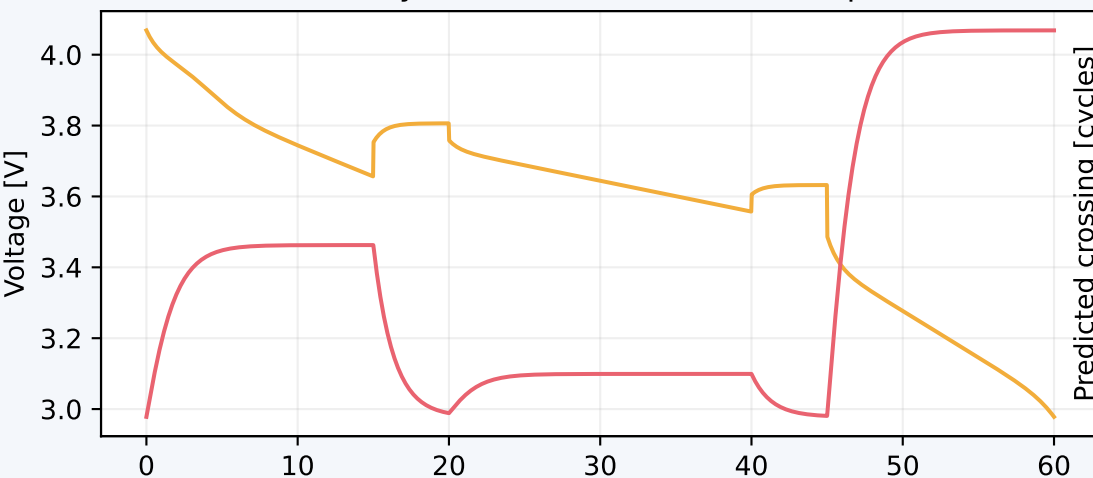
Measured evidence

Two public experimental programs

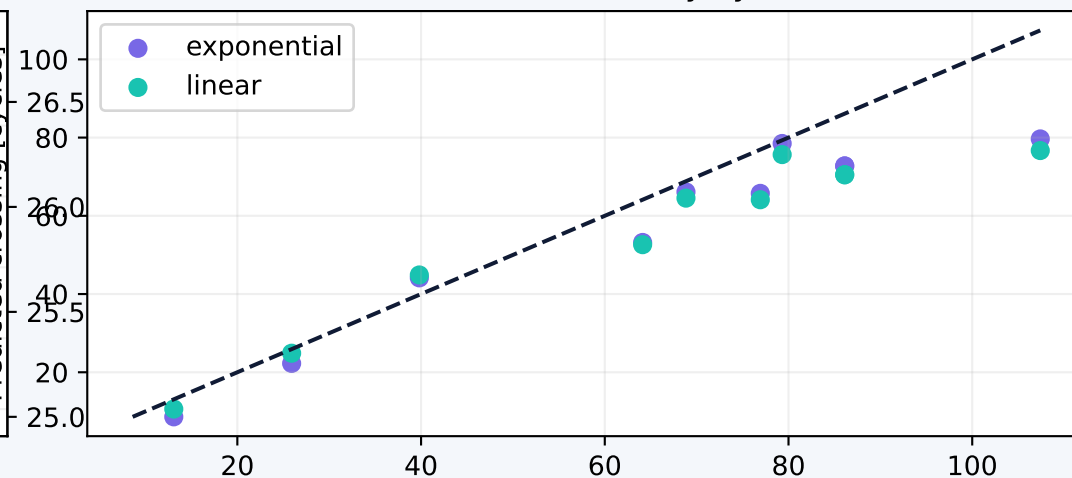
Research outputs

Schemas, tests, notebooks, CSV, JSON, PDF

Level 0 synthetic electro-thermal response



Level 1+ measured early-cycle RUL



478

Calendar rows

0.028

SOH MAE

10

RUL conditions

9.26 cycles

RUL MAE

10 passed

Tests

1 Validate data 2 Simulate physics 3 Fit early cycles 4 Hold out future cycles 5 Report uncertainty